

RUSHIL VENKATESWAR

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EDUCATION

Indian Institute of Technology, Kharagpur <i>Integrated Bachelors and Masters of Technology in Computer Science</i>	Dec 2020 – May 2025 8.71/10
Little Flower School <i>Indian School Certificate Examination (ISC)</i>	Apr 2018 – Apr 2020 96.25%

AWARDS AND ACHIEVEMENTS

- Achieved a peak rating of **1576 (Specialist)** on **Codeforces** and **1868 (Knight)** on **LeetCode** under the handle **rv4102**
- Secured an **AIR 473** in Joint Entrance Exam Advanced 2020 and an **AIR 1176** in Joint Entrance Exam Mains 2020
- Nominated for **Best Bachelor Thesis Award** by Department of Computer Science and Engineering, IIT Kharagpur
- Selected to attend **Optiver Winter School**, a workshop on Quantitative Trading, conducted by IIT Delhi

EXPERIENCE

Sprinklr, India <i>Product Engineering Intern</i> <i>Workflow Automation</i>	May 2024 – July 2024
Objective: Automate after-call workflow graph creation using a chatbot interface and guided-path graph summarization using LLMs	
- Ensured structured output generation from LLMs using Pydantic class definitions, allowing data validation & serialization - Utilized prompt engineering, function calling & query re-asking through Instructor package to generate workflows from queries - Modified open-source packages for compatibility, dockerizing and deploying the tools to reduce workflow creation time by over 60%	
Stanford University <i>Research Intern</i> <i>Maternal and Child Health Monitoring in LMICs</i>	May 2023 – Aug 2023
Objective: Estimate 6 key indicators of health status in low income countries using numeric features derived from satellite imagery	
- Trained a Boosting on Error ensemble to predict 4 level residuals using Random Forest & XGBoost, with MCRMSE of 10.759 - Utilized Dask package on an 8GB dataset for EDA, finally using <code>RF.feature_importances_</code> parameter to perform feature selection - The internship program was conducted as a Kaggle Competition, and our team secured 1st position out of 30 participating teams	
ISRO <i>Research Intern</i> <i>Semantic Segmentation of Remote Sensing Images</i>	Apr 2022 – Dec 2022
Objective: Develop an efficient CNN architecture using Tensorflow for semantic segmentation of geospatial images into 5 classes	
- Modified standard U-Net architecture by adding dilated & depthwise separable convolutions to improve feature extraction - Utilized a novel unified focal loss and created a dataloader for an efficient input pipeline to work with geospatial images	

PROJECTS

Hospital Management System <i>Database Management Systems Lab</i>	Feb 2023 – Mar 2023
Objective: Design and development of an extensible Hospital Management web application for Patient Care and Administration	
- Developed a Python Flask based web application to connect MySQL to a bootstrap front-end coupled with Jinja templates - Implemented user session management using flask-login & provided access control through Python decorator functions	
Linux Shell Development <i>Operating Systems Lab</i>	Jan 2023 – Feb 2023
Objective: Design a pipeline-based shell supporting parallel command execution, detection of malware and file locks	
- Effectively managed process groups and monitored child processes using signal handlers, ensuring synchronized execution - Implemented features such as background execution, pipelining, wildcard handling, and command history navigation	

COMPETITIONS AND CONFERENCES

Inter IIT Tech Meet 12.0 - Gold Medalist <i>Team Captain</i> <i>DevRev – High Prep Event</i>	Sep 2023 – Dec 2023
Objective: Create a low-latency tool-use LLM which matches closed-source LLMs in performance while being cost-effective	
- Created RTaC framework to convert tools to python functions, promoting docstring-reading capabilities in coding-base LLMs - Utilized PEFT to finetune Code Llama 7B & DeepSeek 1.3B on 2500 examples of manually cleaned GPT generated synthetic data - Achieved a competitive performance to GPT-4 at 20% of the cost, supporting dynamic tool addition & mathematical/iterative logic	
Inter IIT Tech Meet 11.0 - Gold Medalist <i>Team Member</i> <i>ISRO – Mid Prep Event</i>	Jan 2023 – Feb 2023
Objective: Create a high-res map of the Moon by stitching image patches generated using super-resolution models on low-res images	
- Proposed a novel GAN-based architecture with separate adversaries for ensuring accurate reconstruction of craters and hills - Developed a Lunar Atlas by correcting coordinates & stitching together individual image patches from Chandrayaan-2 TMC payload - Achieved a competitive SSIM of 0.794 while increasing spatial resolution from 5m/pixel to 30 cm/pixel, a 16x magnification	

RELEVANT COURSEWORK

Theory + Lab: Operating Systems, Computer Networks, Database Management Systems, Computer Organisation & Architecture, Compilers, Software Engineering, Programming & Data Structures, Algorithms-I & II, Distributed Systems
Theory: Deep Learning, Machine Learning, Probability & Statistics, Statistical Inference, Discrete Structures, Linear Algebra

TECHNICAL SKILLS

Languages: C/C++, Python, LaTeX, SQL, Bash, MIPS, Assembly
Skills: Systems Programming, Socket Programming, Data Science, Object Oriented Design
Technologies/Frameworks: Keras, Tensorflow, NumPy, Pandas, Flask, scikit-learn, Git, C++ STL, C pthreads

LEADERSHIP / EXTRACURRICULAR

Student Mentor <i>Students' Welfare Group, IIT Kharagpur</i>	Dec 2022 – Present
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